

Tone 3 Sandhi and Its Interaction with Stress in Mandarin

Abstract

This paper documents Tone 3 sandhi in Mandarin Chinese, a Sino-Tibetan language spoken widely across China and the Chinese diaspora. Data are from elicitation sessions with a native speaker of Mandarin. These data largely confirm the findings of Lin (2007) while adding new evidences of how stress in Mandarin phrases affect the application of Tone 3 sandhi.

Lin describes Tone 3 sandhi in Mandarin as a tonal alternation in which a Tone 3 (214) alternates into a Tone 2 (35) when followed by another Tone 3, as in /laʊ35 ʂu/214 → [laʊ35 ʂu214] ‘mouse’. Tone 3 reduction, another related tonal alternation he describes, is the case where a T3 is reduced or shortened (214 → 21) when they are followed by another tone. The basic domain for Tone 3 sandhi is the disyllabic foot which starts with the smallest morpho-syntactic domain and could extend both forward or backward depending on the phrase structure. For example, phrases like /mi214 laʊ214 ʂu214 xɑʊ214/ alternates to [mi214 laʊ35 ʂu35 xɑʊ214] ‘Mickey Mouse is good’ because the ‘mouse’ is the smallest morpho-syntactic domain in this phrase and Tone 3 sandhi starts with that domain.

The speaker for this paper applies Tone 3 sandhi consistently and naturally in their speech which align with Lin’s findings. The speaker is given the same phrases with different stressed words (e.g. “the small FEMALE horse is good” versus “the small female HORSE is good.”) and is able to demonstrate attempts in not applying Tone 3 sandhi though not always successful. My analysis adds onto Lin’s research by exploring how stress affects Tone 3 sandhi in Mandarin while also following the bounding domain of disyllabic foot and smallest morpho-syntactic domains proposed by Lin.

1. Introduction

Mandarin is the largest language family in Chinese, and is also known as Standard Speaking Chinese. The goal of this paper is to document and analyze tone sandhi and how it interacts with stress in Mandarin with data recorded from a native speaker.

My consultant for this paper is Rongying Zhang, who is a native speaker of Mandarin. She is born and raised in Shenzhen, a southern coastal city in China up until she was 18 years old when she began to study biology at UCLA. She speaks mostly the standard dialect of Mandarin in her daily life while communicating with other people but sometimes speaks with Northeastern China's accent at home as both of her parents come from that region. Growing up, she speaks both Mandarin and English at schools. Despite mostly speaking, reading, and writing in English at UCLA, Rongying continues to use Mandarin when constantly communicating with family and friends from China.

The source for this paper is *The Sounds of Chinese* (2007) by Yen-Hwei Lin. The book includes detailed description of Mandarin phonology and covers major phonological rules of Mandarin including tone sandhi and stress that are focused here. In the following sections, I will first summarize Lin's description of Tone 3 Sandhi and then introduce my own data of how Tone 3 Sandhi interacts with stress.

2. Lin's (2007) Description and Analysis

According to Lin, Mandarin language has four basic tones that differentiate meanings of words: high level (T1), high rising (T2), low falling-rising (T3), and high falling (T4). There is

also a special tone T0 known as the neutral tone that occurs only in non-initial positions of word or phrase and is treated as toneless in notation.

Here are some examples of tones in Mandarin from Lin, demonstrating that tones are crucial part to interpreting words meanings.

(1) Basic tones in Mandarin

Tone Number	Pitch Pattern	Pitch Value	Example
1	High Level	55	[ma] ₅₅ mā ‘mother’ 妈
2	High Rising	35	[ma] ₃₅ má ‘hemp’ 麻
3	Low Falling-Rising	214	[ma] ₂₄₁ mǎ ‘horse’ 马
4	High Falling	51	[ma] ₅₁ mà ‘to scold’ 骂

All four tones are all able to combine with one another. However, the pitch level and contour of tones can change according to contexts. Lin mentions T3 reduction in which T3 are simplified or shortened when they are followed by another tone (p. 196). He gives the following rule and example:

(2) T3 Reduction: 214 → 21 / __ another tone

Example: lǎoyīng ‘eagle’ [lau]214 [jəŋ]55 → [lau]21 [jəŋ]55

An important phonological phenomenon of Mandarin is tone sandhi, particularly the tone 3 sandhi which Lin describes as followed:

(3) Tone 3 Sandhi: T3 → T2 / __ T3 (i.e. 21 + 214 → 35 + 214)

This means that in a sequence of 2 T3s, the first T3 first experience T3 reduction and then changes to T2. Here is an example that illustrates the change:

(4) /ɛjau/214 /kou/214 xiǎogǒu ‘puppy’

[ejau]21 [kou]214 T3 Reduction

[ejau]35 [kou]214 T3 Sandhi

Tone 3 sandhi is more complicated in phrases or sentences that consist of a string of tone 3 words. This means that tone 3 sandhi is not simply dependent on being followed by another T3, below are examples that illustrate some of the possible different surface tone patterns that are produced by the same sequence of T3s, taken from Lin.

(5) mǐ lǎo shǔ hǎo ‘Mickey Mouse is good’

T3 T3 T3 T3 → T3 T2 T2 T3

(6) gǒu yǎo lǎo shǔ ‘The dog bit the mice’

T3 T3 T3 T3 → T2 T3 T2 T3

To understand how tone 3 sandhi is applied in complex words and phrases, Lin introduces the relevant prosodic and morpho-syntactic factors in Mandarin. The basic domain that tone 3 sandhi operates on is constituted by disyllabic foot. The below are examples of how syllables (σ) can be grouped into foot:

- (7) a. $\sigma \sigma \sigma \sigma$ → ($\sigma \sigma$) ($\sigma \sigma$)
- b. $\sigma \sigma \sigma$ → $\sigma (\sigma \sigma)$ → ($\sigma \sigma \sigma$)
- c. $\sigma \sigma \sigma$ → ($\sigma \sigma$) σ → ($\sigma \sigma \sigma$)
- d. ($\sigma \sigma$) ($\sigma \sigma$) → ($\sigma \sigma \sigma \sigma$)

Lin also provides examples of how such grouping is applied in real T3 words sequences:

- (8) a. jiu jiu jiu jiu (jiu jiu) (jiu jiu) ‘nine, nine, nine, nine’
- T3 T3 T3 T3 → (T2 T3) (T2 T3)
- b. jiu jiu jiu jiu jiu (jiu jiu) (jiu jiu) jiu ‘nine, nine, nine, nine, nine’

$$\begin{aligned}
T3 \ T3 \ T3 \ T3 \ T3 &\rightarrow (T2 \ T3) (T2 \ T3) \ T3 \\
&\rightarrow (T2 \ T3) (T2 \ T2 \ T3)
\end{aligned}$$

These examples help establish that disyllabic feet are built from left to right as the last syllable in (8b) is left alone in the first round of grouping before joining the preceding foot to form a three-syllable domain. Tone 3 sandhi is applied after the domains are formed. However, the grouping process in (8) only helps explain (6) but not (5), which requires the initial disyllabic foot grouping of the second and the third syllables to produce the correct result. To account for this kind of grouping, Lin introduces morpho-syntactic domains and asserts that foot building must start with the smallest morpho-syntactic domains, which is *lǎoshǔ* ‘mouse’ in the case of (5). With this, Lin summarizes the rule for foot building for tone 3 sandhi: build the disyllabic foot from left to right and incorporate an unfooted free syllable to its adjacent foot. The process applies first within the smallest domain to the subsequent larger domains step by step, and applies to the smallest domain of phrase to the whole phrase from left to right. The example in (5) can now be explained by the rule as followed, where domains are represented with square brackets:

- (9) [[mǐ [lǎo shǔ]] hǎo] ‘[[Mickey [Mouse]] is good]’
- T3 T3 T3 T3
- (T3 T3) Word: disyllabic foot; T3 sandhi
- (T3 T2 T3) Word: incorporation
- (T3 T2 T2 T3) Phrase: incorporation; T3 sandhi

Lin also states that tone 3 sandhi applies within every domain but is optional when two adjacent T3s belong to different prosodic domains. Expressions with domains crowded toward

the right side are also likely to have multiple patterns compared to the usual single pattern for expressions with domains crowded toward the left side. Alternative additional tone sandhi usually appears in fast or casual speech in which a T3 optionally change into T2 before a T2 that alternated from T3. Below is an example of how an expression can alternate differently based on the forming of domains in fast speech

- (10) [xiǎo [mǔ [lǎo hǔ]]] ‘[small [female [tiger]]]’
- | | |
|---------------|-----------------------------------|
| T3 T3 (T2 T3) | Word: disyllabic foot; T3 sandhi |
| T3 (T3 T2 T3) | Word: incorporation; no T3 sandhi |
| (T2 T3 T2 T3) | Word: incorporation; T3 sandhi |

or Fast Speech Option 1

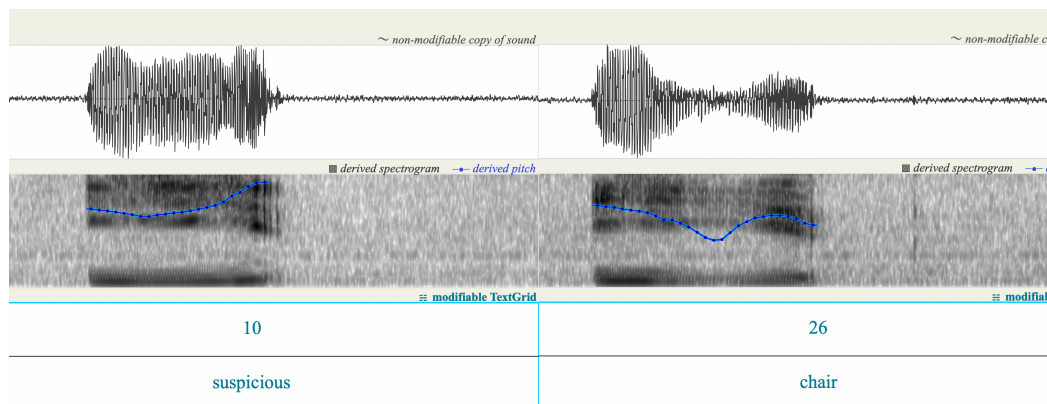
- | | |
|---------------|---------------------------------------|
| T3 (T3 T3 T3) | one prosodic domain for [mǔ [lǎo hǔ]] |
| T3 (T2 T2 T3) | T3 sandhi from left to right |
| (T3 T2 T2 T3) | Word: incorporation; no T3 sandhi |

or Fast Speech Option 2

- | | |
|---------------|---------------------------------------|
| (T3 T3 T3 T3) | one prosodic domain for all syllables |
| (T2 T2 T2 T3) | T3 sandhi from left to right |

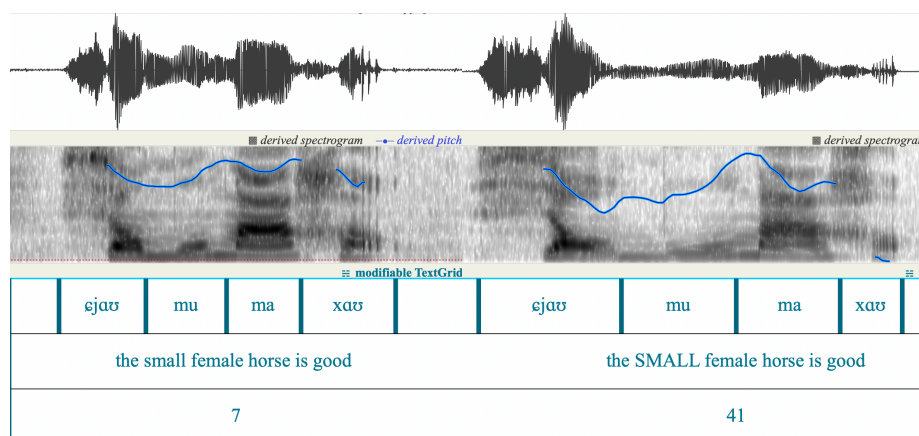
3. Tone 3 Sandhi and Stress in Rongying’s Speech

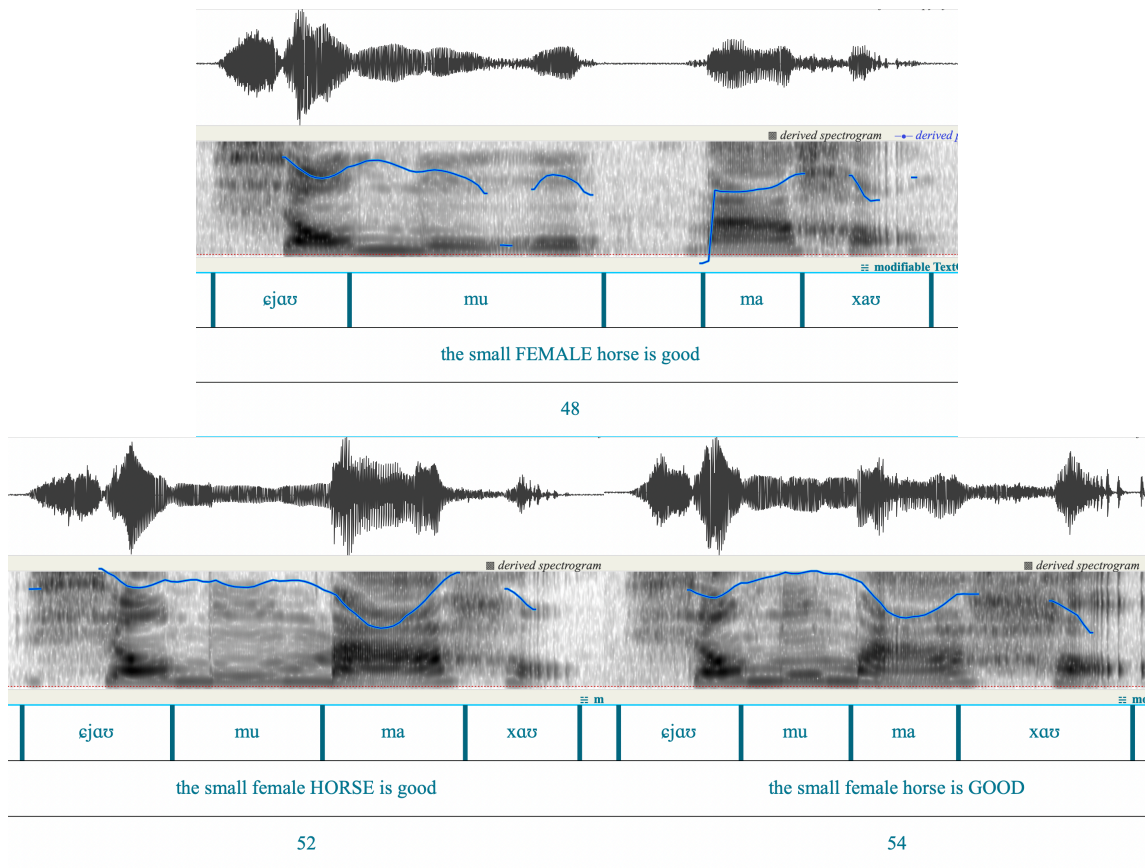
To better understand pitch difference between Tone 2 and Tone 3 in Rongying’s speech for setting a baseline for investigation, we recorded her saying words that only differ in tone. The below example from #10 [ji]35 ‘suspicious’ and #26 [ji]214 ‘chair’ demonstrate the difference between tone 2 and tone 3, with tone 2 showing a consistently rising pitch and tone 3 falling then rising.



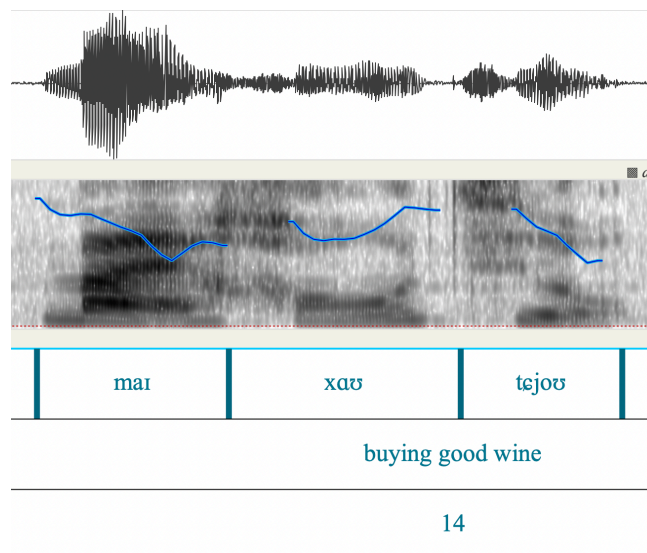
I also elicited words that demonstrate baseline T2 T3, Tone 3 Reduction, and T3 sandhi and their pronunciation aligned with expectations and description from Lin.

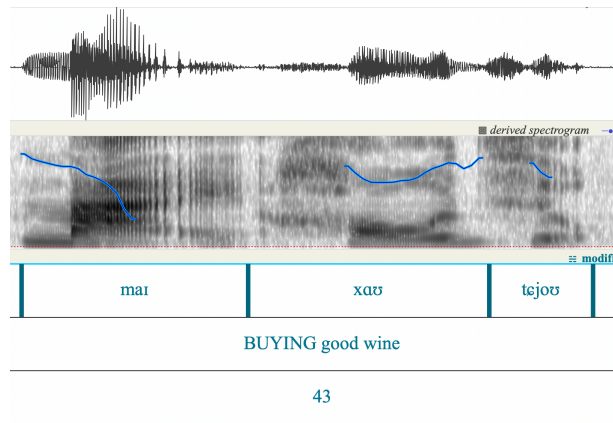
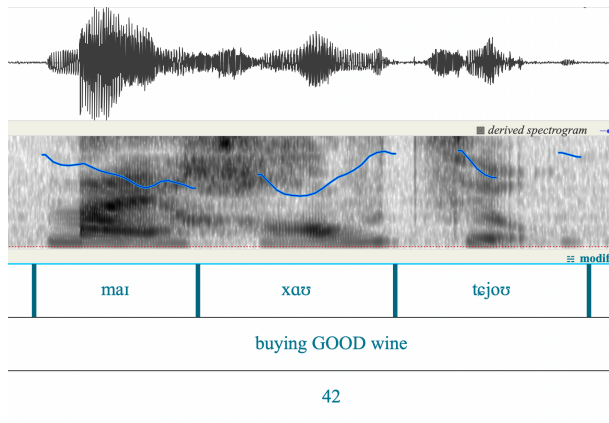
In order to investigate how stress influence Tone 3 sandhi, I ask Rongying to repeat phrases while stressing on different parts of the phrase. A notable example is the phrase #7 “the small female horse is good,” which provides a base case with all Tone 3 words. The most obvious comparison is between #7 and #52 where the word “horse” is stressed. As shown in #52, the pitch of “horse” demonstrate a clear T3 falling and rising pattern rather than the almost flat pattern in #7. #48 is the most unnatural one as my consultant had to take a pause after stress on “female” and showing a clear T3 pattern for it. #41 and #54 both show very similar pattern as the base case #7 which make sense because the proposed T3 sandhi form of #7 didn’t alternate the first word “small” and the last word “good” from T3 to T2.





The change isn't as obvious in phrases like “buying good wine” from #14, #42, and #43. Despite stressing on ‘GOOD’ in #42, the speaker still pronounce the word with Tone 3 sandhi and show no clear difference with #14. However, one can still distinguish the change in pitch range with the [xau] in #42 falling the lowest.





My theory is that

4. Summary and Solution

5. Appendix

Elicitation Word List

#	Transcription	Gloss	Translation
1	[ni]35 [xaʊ]214	‘hello’	你好
2	[ɥy]35 [san]214	‘umbrella’	雨伞
3	[laʊ]35 [ʂu]214	‘mouse’	老鼠
4	[tʂan]35 [lan]35 [kwan]214	‘exhibition hall’	展览馆
5	[ma]35	‘hemp’	麻
6	[laʊ]21 [jəŋ]55	‘eagle’	老鹰
7	[ɕjaʊ]214 [mu]35 [ma]35 [xaʊ]214	‘the small female horse is good’	小母马好
8	[ɕjaʊ]35 [ma]35 [pʰaʊ] 214	‘the small horse is running’	小马跑

9	[mi]214 [laʊ]35 [ʂu]35 [xaʊ]214	‘Mickey Mouse is good’	米老鼠好
10	[ji]35	‘suspicious’	疑
11	[ʂoʊ]35 [liŋ]214	‘chief’	首领
12	[tɕʰi]35 [taʊ]214	‘to pray’	祈祷
13	[ɥy]21 [wən]35	‘language’	语文
14	[mai]214 [xaʊ]35 [tɕjoʊ]214	‘buying good wine’	买好酒
15	[na]35 [ʂuŋ]214 [koʊ]35 [xaʊ] 214	‘which kind of dog is better ’	哪种狗好
16	[tɕʰi]35 [ma]214	‘to ride a horse’	骑马
17	[li]35 [ɕjaŋ]214	‘ideal’	理想
18	[tsʰai]35 [faŋ]214	‘to interview’	采访
19	[ma]214	‘horse’	马
20	[uʊ]35 [ta]214 [laʊ]35 [ʂu]214	‘I hit the mouse’	我打老鼠
21	[pa]35	‘pull out’	拔
22	[ljoʊ]35 [ɕɥə]214	‘to bleed’	流血
23	[pa]214	‘handle’	把
24	[tʂaʊ]214 [ʂoʊ]35 [liŋ]214	‘finding the chief’	找首领
25	[ɥy]35 [fa]214	‘syntax’	语法
26	[ji]214	‘chair’	椅
27	[ta]214 [laʊ]35 [xu]214	‘hitting the tiger’	打老虎
28	[tʂaʊ]214 [tʂan]35 [lan]35 [kwan]214	‘looking for the exhibition hall’	找展览馆
29	[xwei]214	‘destry’	毁
30	[uʊ]35 [ɕjaŋ]214 [mai]35 [tɕjoʊ]214	‘I want to buy wine’	我想买酒
31	[kʰɿ]35 [i]214	‘be capable of’	可以
32	[laʊ]35 [pjaʊ]214 [xən]35 [xaʊ]214	‘the old watch is very good’	老表很好
33	[ɕjoŋ]35 [tʂaŋ]214	‘bear hand’	熊掌

34	[jʊŋ]35 [qœn]214	‘always’	永远
35	[ɕi]35 [tsaʊ]214	‘to shower’	洗澡
36	[tʂʊŋ]35 [li]214	‘to tidy’	整理
37	[xwei]35	‘return’	回
38	[ɥy]21 [ji]51	‘semantics’	语义
39	[tsʰai]35 [faŋ]35 [kaʊ]214	‘interview script’	采访稿
40	[koʊ]214 [xən]35 [xaʊ]35 [jaŋ]214	‘dogs are easy to raise’	狗很好养
41	[ɕjaʊ]214 [mu]35 [ma]35 [xaʊ]214	‘the SMALL female horse is good’	小母马好
42	[mai]214 [xaʊ]35 [tejoʊ]214	‘buying GOOD wine’	买好酒
43	[mai]214 [xaʊ]35 [tejoʊ]214	‘BUYING good wine’	买好酒
44	[laʊ]35 [pjaʊ]214 [xən]35 [xaʊ]214	‘the OLD watch is very good’	老表很好
45	[uʊ]214 [ɕjaŋ]214 [mai]35 [tejoʊ]214	‘I want to buy wine’	我想买酒
46	[laʊ]35 [pjaʊ]214 [xən]214 [xaʊ]214	‘the old watch is VERY good’	老表很好
47	[uʊ]35 [ɕjaŋ]214 [mai]35 [tejoʊ]214	‘I WANT to buy wine’	我想买酒
48	[ɕjaʊ]214 [mu]214 [ma]35 [xaʊ]214	‘the small FEMALE horse is good’	小母马好
49	[mi]214 [laʊ]35 [ʂu]35 [xaʊ]214	‘Mickey Mouse is GOOD’	米老鼠好
50	[uʊ]35 [ɕjaŋ]214 [mai]214 [tejoʊ]214	‘I want to BUY wine’	我想买酒
51	[mi]214 [laʊ]35 [ʂu]214 [xaʊ]214	‘Mickey MOUSE is good’	米老鼠好
52	[ɕjaʊ]214 [mu]35 [ma]214 [xaʊ]214	‘the small female HORSE is good’	小母马好
53	[uʊ]35 [ɕjaŋ]214 [mai]35 [tejoʊ]214	‘I want to buy WINE’	我想买酒
54	[ɕjaʊ]214 [mu]35 [ma]35 [xaʊ]214	‘the small female horse is GOOD’	小母马好
55	[kʰɿ]35 [ni]214 [tʂaʊ]35 [uʊ]214 [mai]35 [pai]35 [wan]214 [mi]35 [fən]214	But you ask me to buy a hundred bowls of rice noodles.	可你找我买百碗米粉

56	[uɔ̌]35 [ɕjaŋ]214 [tɕjɛn]35 [tsɔʊ]214 [ɕjaʊ]35 [ɲjaʊ]214 [weɪ]35 [y]214	I want to pick up the small bird's tail feather.	我想捡走小鸟尾羽
57	[tɕʰɪŋ]35 [ni]35 [tɕaʊ]35 [ljoʊ]214 [tsu]35 [tɕaŋ]214 [maɪ]35 [pa]35 [ɕjaʊ]214 [y]35 [san]214	Please can you buy a small umbrella from team leader Liu?	请你找柳组长买把小雨 伞